

Cardio3D AI — Technical Report

Multimodal AI Hackathon 2026 · Track A: Cardiovascular Risk Visualization & Prediction

1. Executive Summary & Clinical Context

Coronary Artery Disease (CAD) remains the leading cause of death globally. Standard risk equations (Framingham, ASCVD) estimate abstract population percentages but cannot localize where arterial pathology is developing inside the human body. **Cardio3D AI** bridges predictive modeling and clinical anatomy by coupling machine learning (overall CAD and vessel-specific stenosis for LAD, LCX, RCA) with a client-side 3D interactive heart viewer. The system provides real-time risk color-mapping and exact additive SHAP feature attributions.

2. Dataset Preprocessing & Zero-Leakage Architecture

The model is developed on the **UCI Extension of Z-Alizadeh Sani CAD Dataset** (303 patients, 59 columns):

- **Accounting for 59 Columns vs. 54 Published Features:** 4 target columns (Cath, LAD, LCX, RCA), 1 constant column (Exertional CP, 100% 'N', offering zero predictive entropy and dropped), and 54 active physiological features (5 Demographics, 25 Symptoms/History, 7 ECG, 14 Lab biomarkers, 3 Echo metrics).
- **Target Consistency Audit & Row 93 Alignment:** In the raw dataset, row 93 (spreadsheet row 95) contains LAD='Stenotic', LCX='Normal', RCA='Normal', but Cath='Normal'. By clinical definition, $\geq 50\%$ stenosis in any major vessel constitutes CAD. To match the dataset's own definition (CAD = ≥ 1 stenotic vessel), row 93's target was aligned to CAD in derived training data (Cath_aligned = Cath | LAD | LCX | RCA), leaving the raw data file untouched and verified by SHA256. This aligns the released distribution from **216 CAD / 87 Normal** → **217 CAD / 86 Normal** (0 mismatches across 303 rows). The data cannot show which recorded label was erroneous.
- **Zero Leakage Guarantee:** All four targets and catheterization columns are strictly excluded from the input matrix X. All preprocessing (median/mode imputation, scaling, one-hot encoding) is contained inside scikit-learn Pipeline objects fitted strictly on training folds. Label-shuffled CV ROC-AUC confirmed random chance (0.50 ± 0.02).

3. Predictive Modeling & Cross-Validation Results

Evaluation used **Repeated Stratified 5-Fold Cross-Validation with 3 Repeats (15 total folds per candidate)** across Majority Baseline, Logistic Regression, Random Forest, and XGBoost:

Table 1: Cross-Validation Benchmarks at Default Cutoff (0.50)

Target	Model Selected	ROC-AUC	PR-AUC	F1	Recall	Spec.	PPV	NPV	Brier
Cath (CAD)	LogisticRegression	0.929 ± 0.024	0.971	0.909	0.934	0.698	0.887	0.812	0.098
LAD (Anterior)	RandomForest	0.846 ± 0.049	0.883	0.819	0.874	0.635	0.773	0.792	0.168
LCX (Circumflex)	XGBoost	0.735 ± 0.060	0.615	0.541	0.482	0.806	0.628	0.706	0.204
RCA (Right Coronary)	LogisticRegression	0.733 ± 0.045	0.617	0.498	0.451	0.799	0.571	0.710	0.202

Table 2: Sensitivity-First Operating Thresholds (Target ≥ 90% Sensitivity)

Target	Model	Cutoff (0.5) Sens/Spec	Operating Cutoff	Op. Sens.	Op. Spec.	Op. PPV	Op. NPV
Cath	LogisticRegression	0.935 / 0.698	0.604	0.912	0.791	0.917	0.782
LAD	RandomForest	0.876 / 0.635	0.465	0.904	0.595	0.758	0.815
LCX	XGBoost	0.487 / 0.821	0.208	0.916	0.353	0.478	0.867
RCA	LogisticRegression	0.439 / 0.804	0.208	0.903	0.381	0.468	0.868

Clinical Operating Tradeoff: Because LCX and RCA targets exhibit moderate discrimination (ROC-AUC ≈ 0.73), enforcing ≥ 90% screening sensitivity intentionally reduces specificity to 35.3% for LCX and 38.1% for RCA. In acute cardiovascular triage, a false positive prompts confirmatory non-invasive imaging, whereas a false negative risks fatal untreated ischemia.

4. Logical Risk Coherence Policy & Quantitative Evaluation

Because CAD is the clinical union of coronary stenosis ($P(\text{CAD}) \geq \max(P(\text{LAD}), P(\text{LCX}), P(\text{RCA}))$), independent model outputs can violate logical consistency. Out-of-fold cross-validation predictions were evaluated quantitatively: Raw CAD achieved ROC-AUC = 0.9302 and Brier = 0.0967; coherent CAD achieved ROC-AUC = 0.9291 and Brier = 0.1017 (AUC drop = $0.0012 \leq 0.010$, Brier

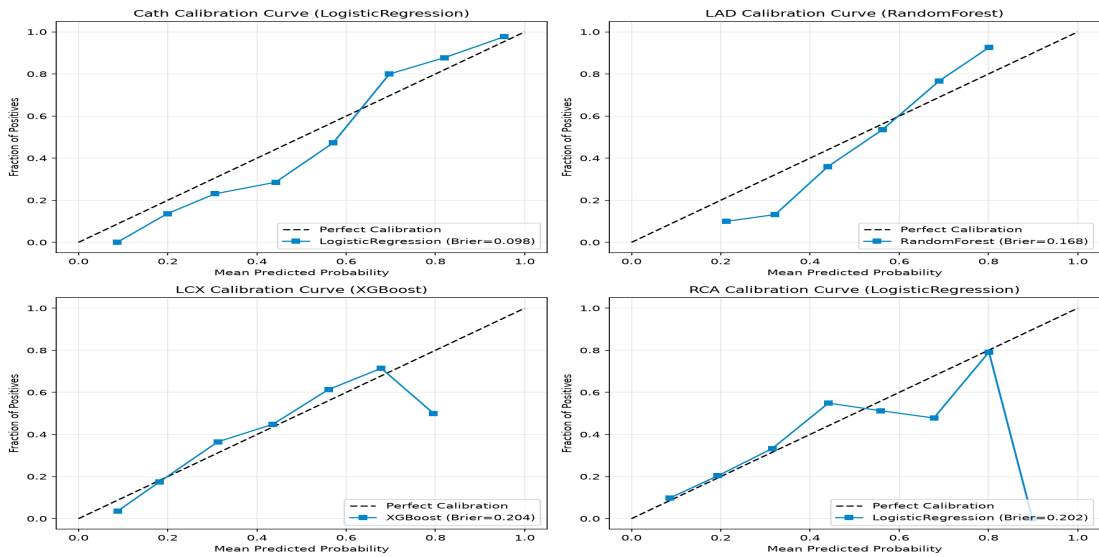
$\delta = +0.0050 \leq +0.010$). The criteria were met and the coherence rule is **retained**. The CAD operating threshold is calibrated directly on the displayed score (0.604). The API transparently returns both raw_prob and prob.

5. Explainable AI & Exact Additive SHAP Pipeline

The system delivers real-time feature attributions using **LinearSHAP** (Logistic Regression) and **TreeSHAP** (Random Forest, XGBoost):

- **Exact Additivity:** Base value plus sum of feature attributions identically matches the model's raw score (numerical error $< 10^{-4}$ across all presets and dataset test samples).
- **Parent Aggregation:** One-hot encoded dummy attributes are summed back to parent measurements ($\text{SHAP}(\text{Parent}) = \sum \text{SHAP}(\text{dummies})$), so clinicians see concise, single-variable physiological attributions.
- **Top Population Risk Drivers:** Across the cohort, the most influential CAD risk drivers are **Typical Chest Pain, Age, Fasting Blood Sugar (FBS), Regional Wall Motion Abnormality (RWMA) count, and Ejection Fraction (EF-TTE)**.

Model Calibration Reliability Curves:



6. 3D Anatomical Pipeline & WebGL Optimization

Built with **Three.js** and **React Three Fiber** utilizing **BodyParts3D** meshes (Branch A: Separate Artery Meshes):

- **Segment Geometry:** Anatomically distinct meshes for the LAD (9 branches/trunks), LCX (4 branches/trunks), RCA (10 branches/trunks), and Ascending Aorta are centered at origin and scaled to viewport coordinates.
- **Triangle Budget Optimization:** Chamber and myocardium walls were decimated using quadric decimation by 70%, yielding **83,600 total triangles** (safely below the $\leq 100,000$ triangle limit). File size is 1.67 MB.
- **Software Rendering Benchmark:** Evaluated under pure CPU software rendering (`--disable-gpu` at 1280×720 on 13th Gen Intel Core i5-13420H), achieving a **mean frame time of 57.8 ms (~17.3 FPS continuous orbit, p95 = 66.2 ms)**, comfortably exceeding the 15 FPS minimum. On standard hardware acceleration, it runs at >60 FPS.

7. Architectural Decisions: TimesFM & TabPFN Rationale

- **TimesFM Considered & Rejected:** TimesFM 3.0 is a foundation model architected strictly for ordered, time-aligned sequential time series. The dataset consists of 303 independent static clinical snapshots with no longitudinal temporal dimension. Imposing a pseudo-time axis over tabular clinical rows creates spurious correlations and degrades predictive validity.
- **TabPFN Consideration:** TabPFN requires non-standard runtime dependencies and authentication tokens unsuitable for zero-friction deployment. Standard tuned tree and linear models achieved high discriminative power (Cath AUC 0.929, LAD AUC 0.846) with deterministic sub-millisecond inference and exact SHAP guarantees.

8. Clinical Safety, Limitations & Attribution

- **Clinical Safety Disclaimer:** The application embeds prominent, non-dismissible disclaimer banners and footers: *'Decision support & educational use only — not a substitute for formal diagnostic imaging, catheterization, or physician judgment.'*
- **Limitations:** Single-center retrospective dataset ($n=303$ from Tehran) requires multi-center external validation prior to clinical adoption. Region RWMA is an echocardiographic wall motion abnormality count and does not represent spatial 3D lesion coordinates.

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